

Node-Network Theory(N2T)

Written in 9 8 月, 2026 by 村长 N2T

目录



Abstract

Keywords

1. Introduction

Preset Parameters (Total: 3)

Observational / Measured Values (Total: 5)

3. Core Viewpoints

4. Derivation Process

4.1 Nodes and Edges: the “Source Code” of the Universe

4.2 Quarks formed by node-connections

4.3 Quarks assemble into elementary particles

4.4 Atoms: combination of atomic-nuclei and electron-clouds

4.5 Molecules: atoms sharing outer-shell grids

4.6 Stars: network-re-construction from hydrogen to iron

4.7 Galaxies: phase-differences between shared-nodes and free-nodes

4.8 Neutron-stars: extreme-compression of outer-shell grids

4.9 Black-holes: ultimate density-state of node-clusters

4.10 The Cosmos: periodic fluctuations of network-wide strain

4.11 Systems: emergence of self-replicating units and intelligence

5 Summary of Derivations: Derivation of 20+ Physical Quantities and Emergent Properties at Each Hierarchy

Derived Quantities

Emergence Manifested Across Hierarchies

6 Open-Problems Remaining Unsolved

7 Quantitative Verification: Two Key Calculations of the Framework

7.1 Natural Cut-off for the Vacuum-Energy Catastrophe

7.2 Geometric Derivation of the Stellar Minimum-Mass

8 Testable Observational Predictions

Prediction 1: Additional small-amplitude periodic oscillations in specific angular scales of the CMB spectrum

Prediction 2: Topological-soliton new-particles with fixed mass windows produced in high-energy particle collisions

Prediction 3: Fixed-spectrum pulsed radiation from topological surface-defects on neutron stars, plus quantitative mass-magnetic-field relation for neutron stars

9 Unified Modeling for All Objects

9.1 Classification of Object Categories

9.2 Unified Modeling of All Objects

10 Conclusions and Outlook

10.1 Summary: From Three Parameters to One Universe

10.2 Outlook: Three Directions for Further Advancement

10.3 Final Open-Ended Question

[References](#)[About the Author](#)[Copyright Statement](#)[Revision History](#)

Abstract

This paper proposes the Node-Network Theory (N2T), a physical model based on discrete structures and stress minimization. The theory holds that the underlying universe is composed of nodes and edges. Nodes generate waves through vibration; phase differences of waves drive stress, and stress drives nodes to rearrange, hierarchically emerging all structures ranging from elementary particles to the macroscopic cosmos. N2T presupposes only three fundamental parameters — total number of nodes N_0 , initial average spacing d_0 , and initial vibration frequency ω_0 . It is calibrated by five observational values (Bohr radius, fine-structure constant, electron mass, total mass of the observable universe, average nucleon mass). After calibration, all other constants emerge naturally from network geometry without additional independent presuppositions. This paper is an open-source thought-experiment, demonstrating a feasible path for deriving cosmic complexity from minimal underlying assumptions. Several open questions remain within the framework, including the geometric mechanism of quantum entanglement, emergence of the arrow of time, and the precise isomorphism between Lie groups and network topology, which await further research and refinement.

Keywords

Node-Network Theory, N2T, vibration, standing wave, travelling wave, stress minimization, six-layer meta-structure

1. Introduction

The author is an IT engineer who proposes the physical model of Node-Network Theory (N2T) based on his own research and practice. In this model, the underlying layer of the universe consists of the basic structures of “nodes + edges”. Various physical phenomena can be deduced using 3 preset parameters and 5 measured values. After completing these deductions, the author further presents a unified modeling approach for both physical objects and human society.

Preset Parameters (Total: 3)

No.	Parameter	Symbol	Value	Description
1	Total number of nodes	N_0	10^{80}	Reference value: number of nucleons within the observable universe
2	Initial average spacing	d_0	10^{-1} m	Derived from the ratio of observable-universe volume to total node count
3	Initial vibration frequency	ω_0	$5.66 \times 10^{18} \text{ rad/s}$	Determined jointly by the fine-structure constant and electron mass

The order-of-magnitude of d_0 is derived from the ratio between the volume of the observable universe and the total number of nodes, approximately 10^{-1} m . This value cannot be directly verified by observation at present, yet it is indirectly constrained by dark-energy density and CMB acoustic oscillations. If $d_0 \ll 0.01 \text{ m}$,

there will be excessive free nodes in vacuum and the dark-energy density will exceed observational data. If $d_0 \gg 1 \text{ m}$, the propagation modes of density waves in the early universe will conflict with CMB observations. Therefore, the current value range of d_0 represents the only self-consistent interval between this framework and astronomical observations.

No.	Observed Quantity	Symbol	Value	Source
1	Bohr radius	r_B	$5.291772 \times 10^{-11} \text{ m}$	Hydrogen atomic spectrum
2	Fine-structure constant	α	$1/137.036$	Quantum electrodynamics
3	Electron mass	m_e	$9.109383 \times 10^{-31} \text{ kg}$	Mass-spectrometry / measurement
4	Total mass of observable universe	M_U	$\approx 10^{53} \text{ kg}$	Cosmological observation
5	Average nucleon mass	m_N	$1.672621 \times 10^{-27} \text{ kg}$	Mass-spectrometry / nuclear-physics measurement (for calibrating N_0)

No additional preset parameters are introduced in the subsequent deduction process.

3. Core Viewpoints

Through research and practice, the author finds that everything in this world consists of objects and their relationships (nodes + edges). An object can be defined as follows:

1. The attributes and functions of an object are determined by its constituent parts and their compositional relationships.
2. The internal structure of an object couples with the external world through emergent properties. Feedback from the outside triggers adjustments to its internal structure, and new structures in turn give rise to new properties.

An object cannot exist without an external environment, because “existence” itself draws a distinction between internal and external. Emergence is not some obscure phenomenon. It simply means that when changes occur inside or outside a subsystem and cross a certain threshold, the subsystem switches its operating mode and exhibits different characteristics.

The entire universe emerges layer-by-step in this manner, starting from the simplest “objects and relationships”.

Emergence occurs universally across different scales of the universe, for example:

- **Water molecules and liquid water:** A single water-molecule has no property of “wetness” or “fluidity”. When large numbers of water-molecules gather at room-temperature and hydrogen-bond networks form and reach a critical scale,

the whole collective exhibits emergent fluidity and surface-tension. These are not new functions of individual water-molecules; rather, scale triggers a collective switch in the interaction modes among molecules.

- **Locusts and locust plagues:** Solitary locusts are green, timid and forage individually. Once population-density exceeds a critical threshold, physical contact frequency among conspecific individuals surges, triggering changes in serotonin levels. Locusts then turn dark-colored, gather in swarms and migrate violently. This is not caused by gene-mutation; external-density parameters trigger a unified shift in the internal state of individuals.
- **Morphable robots:** A robot can rearrange the connection modes of its sub-components (arms, legs, torso), and switch among truck, fighter-jet and combat configurations. Each configuration possesses completely different functional characteristics. These features do not come from the parts themselves, but from the combinatorial topology among parts.
- **Factory normal-mode and emergency-mode:** Under normal orders, one set of production-lines runs according to fixed cycles. When accidents such as power-failure or leakage happen, the system automatically switches to emergency-mode: halting partial production-lines, activating backup power-supplies and readjusting logistics-routes. Factory-equipment itself remains unchanged; what changes are connection-priorities among devices and resource-allocation rules.
- **Normal-state and wartime-mode of a state:** In peacetime, state resources are devoted to people's livelihood, education and infrastructure-construction. Once war breaks out, the state switches to a wartime system: traffic-control, material-allocation, information-propagation and manpower-mobilization are all restructured. The fundamental constituent-elements of the state remain unchanged; what changes are coupling-modes and response-logic among its components.

Boundary note on demonstrative effectiveness of these cases:

Self-organizing systems without designers (water-molecule clusters, locust populations): no preset external control-logic. Macro-characteristics form spontaneously only through local interactions among units. This fully matches the core hypothesis that the universe evolves purely by self-organization without an external creator.

Systems created by intelligent agents (morphable-robots, factory production-lines, state-governance-systems): these contain artificially defined switching-rules and external regulatory subjects. They do not directly prove evolutionary-laws of the natural universe, and serve only as aids for understanding the logic that *changes in topological-structure bring about drastic shifts in macro-characteristics*.

Although these five examples vary greatly in scale, they share the same underlying logic: the macro-characteristics of a system depend on the interaction-modes of its sub-objects that get switched under specific threshold-conditions. Such “mode-switching” is not an accidental phenomenon but fundamental behavior of complex-systems. This is the core principle repeatedly applied in subsequent deductions of the underlying cosmic-structure.

4. Derivation Process

In the following sections, the author attempts to accomplish two goals:

1. Define the most fundamental nodes and edges of this universe;
2. Deduce the properties of all things in the universe starting from their characteristics.

Throughout this process, reasoning proceeds from first principles. Apart from the indispensable preset fundamental parameters (nodes, edges, global-scale parameters), no new parameters are introduced for subsequent emergent phenomena. Although the underlying architecture and emergent processes of this framework may bear similarities to existing mainstream theories, such similarities are merely side-effects of reaching the same phenomena via different paths, not arbitrary patch-work assembly.

Within this framework, whether nodes constitute quarks, atomic nuclei or galaxies, every node executes three minimal rules:

1. Receive: read incoming phase values and stress values from all adjacent edges.
2. Update: calculate its new phase — take the average phase of all neighbors plus quantum-fluctuation noise.
3. Transmit: broadcast its new phase and local stress outward along all connected edges.

4.1 Nodes and Edges: the “Source Code” of the Universe

We may understand it this way: if we keep decomposing objects in the universe, what we finally obtain will no longer be “matter”, but nodes without mass, without size, and without internal structure.

Yet if nodes remain completely inactive, the universe will stay static and unchanged. To “set the universe in motion”, nodes must possess some fundamental activity. That

activity shall be vibration, because only vibration can simultaneously produce the following properties:

- Period and phase — enabling synchronization or phase-offset among nodes;
- Positive and negative values — generating directional attraction or repulsion;
- Amplitude and range — determining interaction strength and effective distance;
- Gradation and propagation — allowing interactions to travel, attenuate and superimpose.

These properties together constitute the full possibilities for interactions between nodes. Without vibration there would be no phase, no waves, no forces, no structures — and therefore no universe. Hence vibration is the sole pre-assigned activity for nodes, and all complex cosmic phenomena originate from it.

(To demand further division beyond this point would sink into nihilism.)

At the birth of the universe, nodes are brought into existence “out of nothing”, and endowed with the basic interaction engine: generating waves via vibration. Attraction and repulsion arise from wave phase (positive / negative), enabling self-organized formation of connections. The minimum stable distance between two nodes equals half-wavelength ($\lambda/2$).

Two forms of waves are distinguished:

- **Standing wave:** locked vibration patterns circulating on closed paths, corresponding to particles such as quarks, electrons and protons.
- **Travelling wave:** vibration patterns propagating freely along open paths, corresponding to photons, radio waves, etc.

In the node-network model, the node-network itself constitutes space; there exists no separate external stage. Node-vibration generates spatial potential. Mathematically this potential has no upper bound and decays following roughly $1/r^2$, never reaching zero. Physically its effective range is determined by node-network density, stress state and phase-coupling strength.

Step 1: Points and vibration emerging “from nothing”

A node may be regarded as an event or a degree-of-freedom in space. In its initial silent state it carries no mass or charge, only potential vibration amplitude $a_i(t)$ and phase $\phi_i(t)$.

Vibration source: each node receives an initial perturbation, like a stone thrown onto still water. Phases and amplitudes may be random, or originate from a unified initial pulse.

Wave generation: once a node vibrates ($a_i \sin(\omega t + \phi_i)$), it radiates waves outwards.

This represents the first act of cosmic creation: transitioning from silence to pulsation, and the birth of information.

Step 2: Wave propagation and attenuation in space

Propagation mechanism: waves travel outward from source points along all possible directions (within discrete networks: along every edge).

Attenuation mechanism: wave intensity decreases with distance. In three-dimensional space it decays as $1/r$ or $1/r^2$; in discrete networks it may be defined as exponential decay $e^{-\lambda L_{ij}}$ with respect to edge-count or geometric distance.

Wave form: a function of space and time $\Psi(x, t)$ carrying both amplitude and phase information.

This forms the information-transfer mechanism: waves replace transmitters; space itself acts as wave medium.

Step 3: Positive-/negative wave phase → attraction and repulsion

Positive / negative refers to the instantaneous phase direction of a wave, i.e. whether $\sin(\omega t + \phi)$ is greater or less than zero. This is not a fixed intrinsic property but dynamically variable.

Interaction rule: after nodes “observe” waves from neighbours, instantaneous phase produces force: same-phase → attraction (tendency to synchronize vibration); opposite-phase → repulsion (tendency for destructive interference).

This mechanism naturally realizes “like repels like, opposites attract”. Here “like” means instantaneous phase-relationship rather than fixed charge labels.

Step 4: Force → formation and evolution of connections

Connection criterion: when sustained attraction arises due to phase-synchronization, nodes draw closer (edge-length decreases).

Connection establishment: when approaching below a threshold value, permanent edges are created, recording phase-difference, distance and coupling strength.

Strengthening / weakening of connections: sustained phase-synchronization increases edge weight; changing phase-relations weaken edges or even break them completely.

“Nodes are defined by their relations”: connections are not pre-existing; they are dynamically generated and maintained through vibrational interactions of nodes.

Stable connections between nodes require three simultaneous conditions: matching of initial spacing and frequency allowing waves to reach each-other; balance between attraction from phase-synchronization and compression-repulsion from waves; vibration-amplitude settling at a stress-minimum state. These three conditions together determine connection rules of the node-network, upon which all higher-level structures are built.

Detailed connection-process explanation:

At the very beginning of the universe, the ability for nodes to form connections and settle at certain distances relates to the relation:

$$d_0 \cdot \omega_0 = c$$

This represents a critical matching condition:

- If $d_0 \cdot \omega_0$ is slightly larger than c : waves arrive with phase offset; locking fails and nodes disperse.
- If $d_0 \cdot \omega_0$ is slightly smaller than c : nodes are pulled together too rapidly, crossing all stable positions and collapsing directly into extremely compact structures (black-holes).
- Only under $d_0 \cdot \omega_0 = c$ can nodes gain just enough speed to cross the first potential-barrier (forming quark-rings), yet stop before the second potential-barrier — yielding hydrogen-atom structures.

Why are hydrogen-atoms formed preferentially? This matching condition permits crossing only up to the first stable standing-wave pattern (3 nodes → quark-ring → proton). More complex multi-nucleus structures such as helium require higher-energy conditions available inside stellar cores.

Step	Mechanism	Condition
1. Wave-tail connection	Wave-tails overlap; phase-locking establishes soft-connections	Same phase and same frequency
2. Nodes draw near	Phase-driven attraction pulls nodes together, counter-acted by compression-derived repulsion	Sufficient approach velocity
3. Crossing potential-barriers	Nodes descend step-wise across standing-wave mode-orders	Velocity determined by $\omega_0 \cdot A_i / d_0$
4. Settle into stable positions	Arrive at certain standing-wave mode-orders; attraction balances repulsion	Stress-minimization condition
5. Hydrogen-atom formation	Nodes settle in the first stable mode, forming proton plus electron-shell	Critical matching: $d_0 \cdot \omega_0 = c$
6. Conditions for heavy-elements	Higher-energy (frequency / amplitude) required to cross additional potential-barriers	Energy supplied by stellar cores

Calculations consist of three components:

- Stress-function: defines current degree of instability (target quantity to be minimized)
- Evolution-rules: defines how nodes respond to stress and move toward stable states
- Convergence-conditions: defines when stability is reached and adjustment may halt

Stress-function $\mathcal{S}$

Total stress contains two terms: phase-stress $\mathcal{S}_\theta$ originating from phase-differences of adjacent nodes; geometric-stress $\mathcal{S}_g$ originating from deviations of edge-length from natural length L_0 .

$$\mathcal{S} = \sum_{\langle i,j \rangle} \left[\alpha(\theta_i - \theta_j - \gamma_{ij})^2 + \beta(L_{ij} - L_0)^2 \right]$$

- θ_i : phase of node i
- γ_{ij} : phase-factor on edge (i, j)
- L_{ij} : actual edge-length
- L_0 : natural length
- α, β : coupling coefficients

Smaller total stress $\mathcal{S}$ corresponds to more stable structures. Evolving toward stability means gradually reducing $\mathcal{S}$.

Evolution-rules

Phase-update for each node:

$$\theta_i^{(t+1)} = \theta_i^{(t)} - \eta_\theta \cdot \frac{\partial S}{\partial \theta_i}$$

Edge-length update for every edge:

$$L_{ij}^{(t+1)} = L_{ij}^{(t)} - \eta_L \cdot \frac{\partial S}{\partial L_{ij}}$$

Where η_θ and η_L represent update step-sizes.

These two formulas express that node-phases and edge-lengths change along directions of steepest descent for total stress – the mathematical formulation for “choosing the locally best available state”.

Convergence-condition for stability

The system reaches stable equilibrium when:

$$\frac{\partial S}{\partial \theta_i} = 0, \quad \frac{\partial S}{\partial L_{ij}} = 0$$

Gradients for all node-phases and edge-lengths vanish; the system sits at a stress-minimum point. For the “shared-central-node quark-ring” structure described earlier, these convergence-conditions naturally yield: phase-differences locked at **120°** among three quark-rings; equal radial-edge lengths; stable hexagonal or quadrilateral connections among outer-perimeter nodes.

Phenomena explainable within this subsection:

- Why the universe exhibits discrete character instead of perfect continuity: underlying discrete node-network.

- Why information-propagation speed is finite: wave-transmission along edges consumes time (delay τ_{ij}).
- Origins of attraction and repulsion: stress generated by wave-phase-differences.
- Three-dimensional space: global statistical geometric distribution of all edge-lengths L_{ij} gives rise to spatial scale and coordinates.
- Time-dimension: iterative sequence determined by edge-propagation-delay τ_{ij} yields unidirectional temporal evolution.

4.2 Quarks formed by node-connections

Within this framework quarks are no longer treated as fundamental particles; they represent stable topological solitons inside vibrational networks, namely phase-locked rings.

Where does the number “three” for quarks originate? To maintain structural stability against fluctuations, closed-loop standing-wave conditions must hold: after one full circuit around a closed loop, wave-phase must return to its starting value. Two nodes can only form open-chains (open strings) incapable of closed-loop configurations. Three nodes may build triangular closed-loops supporting stable standing-wave modes.

“Color charge” may be naturally interpreted as relative phase-differences of the three nodes: Red = 0° , Green = 120° , Blue = 240° . Summing to 360° ($0 + 120 + 240 = 360^\circ$) yields zero net-wave-phase (“color-neutral”), matching SU(3) symmetry for quark color-charge. Equilateral-triangle closed-loops with fixed 120° phase-differences among nodes realize third-order cyclic-group transformations. When extended into three-dimensional phase-space, the

group-structure is strictly isomorphic to the SU(3) gauge-group of quantum-chromodynamics. The three phase-values correspond to red-green-blue color-charges; summing-to- 360° realizes color-neutral states. This reproduces quark-color-symmetry without artificial numerical fitting.

Geometric constraints for triangular-structures:

1. Equal edge-lengths (equilateral triangle): otherwise differing wave-propagation-times on edges produce phase-drift and structural disintegration. The vibration-attraction mechanism automatically adjusts edge-lengths.
2. Phase-locking: three-node phases maintain fixed **120°** offsets; each node adjusts in real-time according to neighbours such that their sum always equals **360°** .
3. Collective vibrational-modes: collective oscillation of the whole triangle manifests as “mass”. Wave-propagation along edges forms discrete versions of the gluon-field: elastic-stress maintaining structural stability.

Particle identities are determined by vibrational-modes:

- Lower collective-vibration-frequency (longer wavelength): light quarks (up / down).
- Higher collective-vibration-frequency (shorter wavelength): heavy quarks (charm / top).
- Asymmetric vibrational-modes (unequal amplitudes for three nodes): excited-state resonance-particles.

Therefore six quark-flavors do not need to be presupposed; different intrinsic-modes of triangular closed-loops naturally correspond to distinct quark flavors.

Hadrons assembled from two- or three-quarks:

- Mesons (quark-antiquark pairs): two phase-opposite triangular-rings connected via one “gluon-edge”; total phase-difference equals zero.
- Baryons (three-quarks): three triangular-rings nested into larger triangular closed-structures, overall phase-sum 360° (color-neutral).

Emergence of color-confinement: any attempt to pull-out a single node breaks closed-loop standing-wave-conditions. The system immediately generates fresh nodes (wave-pairs) at fracture-points for compensation. Calculation from loop-stress-equations shows energy required for separating two quarks grows linearly with distance, matching QCD confinement energy-scales.

Mass-generation-mechanisms:

1. Base-mass: fundamental-frequency ω of triangular / hexagonal closed-loops corresponds to intrinsic base-particle-mass.
2. Higgs-strain correction-mass: deviations ΔL_0 of loop-edge-lengths from equilibrium-length produce geometric-stress-coupling contributing mass-correction-terms.
3. Topological-leakage tiny-mass: open-chains or solitons without closed-loops produce only second-order weak-coupling leakage-mass, accounting for the extremely light character of neutrinos.

These three mass-categories represent different terms within one unified stress-function and possess no mutually-disconnected independent mass-origins. Nodes themselves carry no mass. At micro-scale mass originates from Higgs-strain caused by edge-length-deviations away from equilibrium-length; macro-scopically this

strain statistically manifests as spatial distribution of node-density. Both descriptions represent equivalent expressions of the same mechanism at different scales.

Stress level	Corresponding phenomenon	Observed manifestation
Extremely high	Interior of atomic-nuclei	Strong-interaction, nuclear-energy
High	Atoms, molecules, stars	Ordinary matter (4.86)
Intermediate	Galactic halos	Dark matter ($\sim$ 27)
Extremely low	Interstellar-space, dispersed vacuum-nodes	Dark energy ($\sim$ 68)

Phenomena explainable in this subsection:

- Why quarks possess three “colors”: **120°** phase-differences among triangular-loop nodes.
- Why quarks are confined: triangular closed-loops cannot break apart without generating new nodes.
- Why quarks exhibit different flavors: different intrinsic vibrational-modes of triangular-loops.
- How hadrons are assembled from quarks: nesting and edge-sharing of triangular-rings.

4.3 Quarks assemble into elementary particles

Now composite triangular-ring complexes are assembled into observable elementary particles: protons, neutrons, electrons and neutrinos.

Proton (three-quark, uud): three triangular-rings (up, up, down-quark) stack by sharing one central-node forming tetrahedral-geometry (4 nodes, 6 edges). Each triangular-ring occupies one face; overall color-neutral. Within network-description: three triangular-rings share one central-node forming a trefoil-knot topology.

Neutron (three-quark, udd): two triangular-rings (up, down) share one node; a third triangular-ring (down-quark) attaches to the side via gluon-edge. Its overall vibrational-frequency is slightly higher than that of the proton due to extra vibrational-mass-contribution from the down-quark.

Electron (hexagonal standing-wave): the electron represents a specific stable standing-wave-mode within the network formed by vibration-resonance confined on closed-loops bounded by six edges. Nodes merely serve as support-structures for waves, comparable to fixed-points on a vibrating string rather than constituents of the wave itself. On loops bounded by six edges, standing-wave-modes satisfying resonance-conditions (wavelength equals integer-fraction of loop-perimeter) manifest as electrons. This standing-wave carries charge (phase-stress S_θ perturbation), spin (handedness of loop-vibration), and mass (coupling to Higgs-strain-field).

Neutrinos: Neutrinos are not built from fixed-counts of nodes; their geometry differs fundamentally from quark triangular-loops and electron hexagonal-loops. They barely

couple to Higgs-strain-fields, meaning they cannot form complete strain-coupling closed-circuits. Two candidate-configurations:

- Single-node soliton: isolated node weakly connected by one weak-coupling edge, no closed-loop; mass comes only from second-order topological-leakage and is extremely small.
- Two-node open-chain: two-nodes linked by weak-coupling-edge forming open non-closed structures capable of flipping along the time-direction (Majorana-like properties) with almost zero Higgs-field-coupling.

Neutrinos carry no charge and self-balance their phase-stress; they may adopt simpler open-chain or soliton-geometries without requiring compensation-closed-loops. Essentially neutrinos represent topological-defects propagating through weakly-coupled networks with nearly-absent closed-loops. Their tiny masses originate from transient strain-perturbations of regions they traverse. Single-node-solitons and two-node open-chains constitute base topologies; phase-mixing between these two topologies generates flavor-eigenstates for three neutrino-generations. Topological phase-differences along propagation-paths produce neutrino-oscillation-phenomena.

Phenomena explainable in this subsection:

- Distinction between protons and neutrons: different arrangements (uud versus udd) for triangular-rings.
- Origins of electron-charge, mass and spin: phase-stress, strain-coupling and handed-topology of hexagonal-loops.
- Why neutrinos are ultra-light, uncharged and hard-to-detect: open-chain / soliton-configurations absent from closed-strain-loops.

- Why protons are slightly lighter than neutrons: additional vibrational-mode-contribution from the neutron's down-quark.

4.4 Atoms: combination of atomic-nuclei and electron-clouds

Atomic-nuclei (node-clusters built from protons and neutrons) couple via radial-edges to outer-shell hexagonal-grids (support-structures for electron standing-waves), forming atoms.

Geometric-networks of multi-nucleon atomic-nuclei:

- Deuteron (2-nucleon): tetrahedra (proton + neutron) share one edge to build octahedral-geometry.
- Helium-4 nucleus (4-nucleon): four tetrahedra share vertices to form icosahedral-geometry of exceptionally high symmetry, explaining the extraordinary stability of helium-4.
- Heavy-nuclei (iron, uranium etc.): large-quantities of tetrahedral-units stack fractally, every closed-loop satisfying zero-total-phase-sum constraints (generalization of color-confinement for nuclear-scale).

Coupling with phase-and-geometry dual-stress-networks:

- Phase-stress drives intra-nuclear-structures: triangular-rings maintain **120°** internal phase-differences; inter-ring phase-gradients arise from shared-node phase-variations. External electric-fields modify intra-nuclear phase-distributions producing polarization-responses.
- Geometric-stress determines nuclear-size and shape: edge-length L_{ij} governed by geometric-stress S_g ; strong-interaction acts as tension along

edges yielding nuclear-radius formula $R \approx 1.2A^{1/3} \text{ fm}$.

- Cross-terms generate nuclear-shell-structure: combined action of phase-stress and geometric-stress produces periodic fluctuations in node-density versus radius. Stable standing-wave-modes correspond to nuclear magic-numbers: 2, 8, 20, 28, 50, 82, 126.

Total stress-function covering sub-nuclear to nuclear-scales:

$$S_{\text{total}} = \alpha S_{\theta} + \beta S_g + \gamma S_{\text{hadron}}$$

- S_{θ} : squared phase-differences among nodes (electromagnetism plus quark-color-constraints)
- S_g : squared deviations of edge-lengths (gravitation plus nuclear-tension)
- S_{hadron} : phase-sum-deviations for triangular-rings (enforcing color-neutrality)

Coupling from atomic-nucleus toward atomic-orbitals:

Treating an atomic-nucleus as one aggregate node (mass M , charge Z), it couples electrically to surrounding electron-nodes (phase-rings). Electrons form phase-standing-wave orbits around nuclei. Orbital-radius emerges as Bohr-radius from equilibrium between phase-stress S_{θ} and geometric-stress S_g .

Electron energy-levels: standing-wave eigen-modes of outer-shell grids

Outer-shell hexagonal-grids surrounding atomic-nuclei are not flat homogeneous-sheets but closed polyhedral-grids centered on atomic-nuclei, analogous to honeycomb-meshes on spherical-surfaces. Electron standing-waves

correspond to eigen-vibrational-modes on these grids. Different nuclear-node-counts N_{nuc} produce distinct outer-shell-grid topologies:

- Hydrogen ($Z = 1$): cubic outer-shell-grid (8 vertices), standing-wave-modes covering six faces.
- Helium ($Z = 2$): regular-dodecahedral outer-shell-grid (20 vertices), standing-wave-modes covering twelve faces.
- Lithium ($Z = 3$): truncated-icosahedral outer-shell-grid (60 vertices), standing-wave-modes covering twelve pentagonal and twenty-hexagonal faces.

Standing-waves on spherical-grids possess discrete frequencies just like guitar-strings. These modes feature radial and angular nodal-planes:

- K-shell ($n = 1$): ground-state mode without radial nodal-planes, lowest-frequency.
- L-shell ($n = 2$): first-excited-state with one radial nodal-plane, higher-frequency.
- M-shell ($n = 3$): second-excited-state with two radial nodal-planes, further-increased frequency.

Energies follow approximate relation $E_n \propto 1/n^2$, reproducing Rydberg-formula for hydrogen-spectra purely as geometric consequence of spherical-grid standing-waves, without solving the Schrödinger-equation.

Angular nodal-planes correspond to orbital-types:

- s-orbital: fully-symmetric mode without angular nodal-planes.
- p-orbital: dipole-mode possessing one angular nodal-plane (phase-flip around one axis).

- d-orbital: quadrupole-mode possessing two angular nodal-planes.
- f-orbital: octupole-mode possessing three angular nodal-planes.

These modes correspond exactly to angular-quantum-number l in standard-physics but expressed in geometric-language. As nuclear-charge Z increases, outer-shell-grids expand sequentially; whenever new nuclear-nodes complete fuller polyhedral-closures, frequency-spectra exhibit jumps or stable-valleys, giving rise to ionization-energy-peaks and noble-gas stable-configurations of the periodic-table.

Concept from standard-model	Correspondence within this framework
Principal quantum-number n	Radial nodal-planes of standing-waves on outer-shell-grids
Azimuthal quantum-number l	Angular-nodal-plane symmetry of outer-shell-grid standing-waves
Magnetic quantum-number m	Spatial-orientation of standing-wave-modes within three-dimensional-space
Energy-level spacing $\propto 1/n^2$	Eigen-frequency-spectrum for spherical-grid standing-waves
Ionization-energy periodic-trends	Sequential-expansion of outer-shell-grid-topology versus increasing Z
Electron-cloud probability-density distribution	Vibration-amplitude-distribution of standing-waves across grid-nodes

Therefore quantization of electron-energy-levels is not an artificially-imposed physical-law but natural discreteness of standing-wave-frequencies on closed spherical-grids. This yields a geometric-answer to the question: “Why can electrons only occupy discrete orbits?”.

Phenomena explainable in this subsection:

- Stability of atomic-nuclei: topological-constraint requiring zero-sum phase for closed-loops.
- Periodic-table of elements: increasing $Z \rightarrow$ changing outer-shell-grid-topology $\rightarrow$ distinct electron standing-wave-modes.
- Atomic-radii: circum-sphere-radii of outer-shell hexagonal-grids.
- Spectral-lines: transitions of electron standing-waves between different orbital-modes.

4.5 Molecules: atoms sharing outer-shell grids

When two atoms approach each-other, their outer-shell hexagonal-grids couple through edge-sharing and form molecules.

Covalent-bonds = edge-sharing between outer-shell-grids

Condition for edge-sharing: phase-matching on shared-edges (zero phase-difference across shared-edge).

Consequence of edge-sharing: atomic-nucleus vibration-phases couple through shared-edges into larger resonant-systems, yielding collective molecular vibrational-modes.

Bond-strength proportional to quantity and topology of shared-edges:

- Single-bond: share 1 edge (connecting two-nodes).
- Double-bond: share 2 edges (connecting four-nodes).
- Triple-bond: share 3 edges (connecting six-nodes).

Bond-strength scales with stress-coupling-coefficient determined by shared-edge-length and phase-matching-quality.

Molecular-orbitals = standing-wave-modes on merged outer-shell-grids

After outer-shell-grids share edges, the combined outer-shell becomes one larger hexagonal-network.

- Bonding-orbital: in-phase superposition → electron-density concentrates around shared-edge-regions (“chemical-bond”); total phase-stress decreases and energy is released.
- Antibonding-orbital: anti-phase superposition → electron-density avoids shared-edge-regions (“antibond”); total phase-stress rises and energy-input is required.

Bond-strength is proportional to number of shared-edges and phase-matching-coupling-coefficient. Total-stress-differences allow direct computation of covalent-bond-energies and minimum-stress molecular-geometries (for example the 104.5° bond-angle for water-molecules).

Electronegativity = stress-sensitivity of outer-shell-grids

High-electronegativity elements such as oxygen and fluorine possess smaller outer-shell-grids (fewer vertices), high sensitivity toward stress-perturbations on shared-edges and tend to attract more electron-density. Low-electronegativity elements such as sodium and potassium feature larger outer-shell-grids with many vertices, lower stress-sensitivity and tend to donate electrons. Within this framework electronegativity equals ratio: $N_{\text{nuc}} / N_{\text{shell-vertex}}$. For fluorine / oxygen this ratio is large giving high electronegativity; sodium / potassium yield small-ratios favouring electron-donation. Calculations using grid-vertex-counts reproduce trends from

Pauling-electronegativity-scale, confirming this ratio possesses quantitative explanatory-power for chemistry.

Chemical-reactions = reconstruction of outer-shell-grids

1. Two outer-shell-grids couple via shared-edges.
2. Following stress-minimization-principles some edges break and new edges form through redistribution.
3. Electron standing-wave-modes transform from atomic-orbitals into molecular-orbitals.
4. Energy-release or absorption corresponds to variation of total-network-stress S_{total} .

Phenomena explainable in this subsection:

- Nature of covalent-bonds: edge-sharing of outer-shell-grids.
- Bond-order: count of shared-edges (single / double / triple-bonds).
- Molecular-orbitals: standing-wave-modes on merged outer-shell-grids.
- Electronegativity: ratio between nuclear-node-count and outer-shell-vertex-count.
- Chemical-reactions: reconstruction operations performed on outer-shell-grids.
- Molecular-geometry (e.g. bond-angle 104.5° for H_2O): minimum-stress configurations of shared-edges inside three-dimensional-grids.

4.6 Stars: network-re-construction from hydrogen to iron

Stars represent cosmic-scale stress-release-machineries. High-pressure and high-temperature force outer-shell-grid-sharing among atomic-nuclei and trigger

chained network-re-construction that synthesizes heavy-elements from light-species.

Densification: erasure of individual-atom boundaries

At low-density each hydrogen-atom possesses its independent cubic outer-shell-grid (8-vertices). Under high-density outer-shell-grids spatially overlap; nodes originally belonging exclusively to single-atoms become shared among multiple atomic-nuclei. Critical-condition: when inter-nuclear-spacing shrinks below outer-shell-hexagonal-grid-diameter, the notion of individual-atoms loses validity.

Resonance-amplification: superposition of stresses

Within stellar-cores at high-density large-quantities of atomic-nuclear outer-shell-grids overlap creating shared-grid-regions. Vibrational-waves from multiple atomic-nuclei coherently superpose inside shared-grids and locally amplify standing-wave-amplitudes. Amplitude scales as $\sqrt{N}$ while stress scales as N ($\sigma \propto A^2$). When number of participating nodes within shared-regions reaches local-threshold $N_{\text{shared}} \approx 1/\alpha \approx 137$, local-stress exceeds combination-threshold triggering micro-fusion-seed-events.

Macro-scale conditions for sustained fusion inside stars:

- Sufficient volume: supply enormous-quantities ($\approx 10^{56}$) of simultaneous resonance-units.
- Sufficient density: maintain persistent overlapping shared-grids preventing their dissipation.
- Sufficient temperature: supply initial vibrational-amplitude enabling resonance-amplification.
- Gravitational-confinement: trap reactive material within finite-volume preventing dispersal of shared-grids via expansion.

Only when all macro-conditions hold simultaneously do micro-resonance-events concatenate into sustained fusion-chains. That explains why the Sun must achieve a certain volume-threshold to ignite. Stellar-fusion is not merely a process for 137 atomic-nuclei; it represents collective-co-resonance of 10^{56} atomic-nuclei on astronomical-scales.

Network-re-construction: nuclear-fusion

1. Two hydrogen-nuclei couple phase-locked through shared outer-shell-grids.
2. Former boundary-edges are internalized and transformed into intra-nuclear-edges.
3. Merge two atomic-nuclear-node-sets plus shared-grid-nodes into one larger node-cluster.
4. Output: deuterium-framework.
5. Iterate: hydrogen $\rightarrow$ deuterium $\rightarrow$ helium-3 $\rightarrow$ helium-4 (tetrahedral-structure) $\rightarrow$ carbon $\rightarrow$ oxygen $\rightarrow$... $\rightarrow$ iron.

Energy-release

Every internalization-step reduces total-network-stress S_{total} . Surplus stress-energy radiates outward in form of vibrational-waves: fusion-energy. Iron represents termination-point: ratio $R = \sigma_{\text{inner}} / \sigma_{\text{outer}}$ for iron-atomic-nuclei reaches global-minimum and its network configuration achieves maximum-stability.

Layered-structure inside stars

Hydrogen-atoms exhibit systematic state-variations with depth inside stars, giving-rise to stellar-layering:

- Core (high-T high-P): huge vibrational-amplitudes; outer-shell-grids are forced to share; phase-stress dominates; many atomic-nuclei form resonant

standing-waves triggering fusion.

- Radiative-zone (intermediate-T intermediate-P): amplitudes decrease; resonance-units dissolve; energy propagates outward as travelling-waves via radiation-transport.
- Convective-zone (low-T low-P): amplitudes further-drop; geometric-stress (density-gradients) dominates; node-clusters drift collectively producing convection-currents.

No hard-boundaries separate these three-zones. They represent continuous phase-transitions of the same vibration-stress-mode-switching-mechanism under varying depth-conditions. Macro-scale stellar-layering emerges as statistical manifestation of micro-state-switches for hydrogen-atoms with depth.

Phenomena explainable in this subsection:

- Why stars shine: stress-waves released through network-re-construction (fusion-energy).
- Element-synthesis-sequence: hydrogen $\rightarrow$ helium $\rightarrow$ carbon $\rightarrow$ oxygen $\rightarrow$... $\rightarrow$ iron via sequential sharing-and-re-construction.
- Why iron marks the endpoint of stellar-nucleosynthesis: iron-nucleus reaches minimal-network-stress-state.
- Core-temperature of Sun (~ 15 million K): jointly determined by resonance-condition $N_{\text{shared}} \approx 137$ plus network-non-linear-corrections.

4.7 Galaxies: phase-differences between shared-nodes and free-nodes

At galactic-scales phase-differences between visible-matter (shared-nodes) and dark-matter (free-nodes) produce long-range geometric-stress that shapes

galactic-morphology and kinematics.

Dark-matter = free-nodes within network that fail to form standing-waves

Some nodes fail to integrate into stable standing-wave-configurations

(hexagonal-loops) due to mismatches in geometric-stress S_g or phase-stress S_θ .

These solitary-nodes (solitons, open-chains, small-clusters) do not phase-synchronize with outer-shell-hexagonal-grids and couple extremely weakly to electromagnetic-fields (photons). They interact with the global-network only via geometric-stress S_g , modifying edge-lengths and frequencies without emitting-light.

Galactic rotation-curves

Shared-nodes (visible-matter) concentrate toward galactic-centres forming high-density node-clusters. Free-nodes (dark-matter) populate extended low-density outer-halos. Long-range geometric-stress exerted by free-nodes supplies additional centripetal-drag upon visible-matter, flattening galactic-rotation-curves against distance.

Galaxy-formation

Initial node-density-perturbations grow under geometric-stress-driven gravitational-instability. Shared-nodes aggregate into dense galactic-cores; free-nodes build outer-halos. Phase-difference-distributions between shared-nodes and free-nodes determine galaxy-morphology: spiral, elliptical, irregular-galaxies. Large-scale cosmic-structures (galaxy-clusters, cosmic-filaments) arise via hierarchical-merging of node-clusters.

Phenomena explainable in this subsection:

- Dark-matter: free-nodes not incorporated into standing-wave-configurations.
- Flat galactic-rotation-curves: long-range geometric-stress from free-node-halos.

- Diversity of galactic-morphologies: distributions of phase-differences between shared-nodes and free-nodes.
- Large-scale cosmic-structures (clusters, filaments): hierarchical merging of node-clusters.

4.8 Neutron-stars: extreme-compression of outer-shell grids

When massive-stellar-cores collapse, outer-shell-grids get compressed to their absolute-limits maximizing edge-sharing and yielding neutron-stars.

Extreme-compression shatters outer-shell-grids: inter-nuclear-spacing drops below closed-shell-radius for outer-shell-grids. Hexagonal-grids supporting electron-standing-waves are crushed completely and edges lose independent-existence. Electrons combine with protons via weak-edge “identity-rewrite”-operations to form neutrons. Node-clusters come into direct contact. Inside neutron-stars formerly indirect interactions mediated by outer-shell-grids vanish; neutron-node-clusters interconnect directly via edges into one huge continuous-network built-from neutron-nodes (each neutron corresponds to seven elementary-nodes).

Pulsar-signals

Neutron-stars rotate at high-speeds. Local topological-defects exist at their surfaces. As rotation sweeps defects across our line-of-sight they emit coherently-phased wave-beams producing pulsed-signals. Pulse-period couples to intrinsic vibrational-frequency of neutron-star-networks.

Ultra-strong magnetic-fields

Spin-directions (twist-orientations of loops) of large-quantities of neutron-nodes

become frozen into highly-ordered alignment during outer-shell-grid-collapse. Uniform orientation of phase-factors along huge-quantities of edges generates ultra-strong magnetic-fields reaching up to 10^{11} T.

Phenomena explainable in this subsection:

- Density-limits for neutron-stars: complete collapse of outer-shell-grids.
- Pulsar-signals: periodic phase-beams originating from rotating topological-defects on node-clusters.
- Ultra-strong magnetic-fields: frozen highly-ordered spin-orientations.
- Distinction between neutron-stars and ordinary-matter: presence or absence of intact outer-shell-grids.

4.9 Black-holes: ultimate density-state of node-clusters

When node-cluster-density exceeds critical-value edges are shared to absolute-limits producing one monolithic giant-node-cluster: a black-hole.

Density surpassing critical-value $\rightarrow$ maximal-edge-sharing: above approximately three-solar-masses geometric-stress S_g overwhelms phase-stress S_θ .

Node-positions tend toward coincidence; edge-lengths shrink toward zero. The network enters globally phase-synchronized state without local-differentiable features.

Event-horizon = propagation-limit for network-information

Within standard-models the horizon represents a boundary beyond which light cannot escape. In this framework the horizon is defined as the location where wave-speed

v_w along edges falls to zero. Phase-perturbations can no longer propagate outwards; no functional-edges connect exterior-nodes with the interior giant-node-cluster.

Singularity elimination

Mathematical-singularity (infinite-density) from general-relativity disappears in this framework. Once node-clusters reach saturation-density-threshold further shrinkage ceases. The system enters static fully-phase-synchronized topological-saturated-state with no further stress available for release.

Hawking-radiation = stress-leakage across boundary-edges

Near-horizon phase-noise transiently fractures some edges. One node-fragment falls inward into giant-cluster while the other fragment escapes carrying small-quantities of stress-energy: this constitutes Hawking-radiation, a second-order-effect whose radiation-intensity scales with boundary-curvature of giant-node-cluster.

Topological-saturation of black-holes is mathematically equivalent to Schwarzschild-radius-solutions from general-relativity once node-cluster-density hits saturation-threshold, eliminating mathematical-singularities and re-producing event-horizon-physics via network-geometric-boundary-conditions.

Phenomena explainable in this subsection:

- Why black-holes emit no-light: global phase-synchronization suppresses local standing-wave-modes.
- Event-horizon: boundary where wave-speed drops to zero.
- Non-existence of physical-singularities: topological-saturated state for node-clusters.
- Hawking-radiation: edge-fracture-and-escape-mechanisms near event-horizon.

4.10 The Cosmos: periodic fluctuations of network-wide strain

Zooming to cosmic-scale, the entire universe itself represents one giant-node-network. Hubble-expansion, dark-energy and cosmic-microwave-background are macroscopic manifestations of network-average strain.

Network-average edge-length acts as cosmic scale-factor: cosmic-expansion corresponds to growth of network-average-edge-length: $a(t) = L(t)/L_0$. Hubble-constant H_0 equals relative growth-rate of average-edge-length $\dot{L}/L$.

Dark-energy = global residual of network-wide phase-strain

Phase-stress S_θ cannot be fully cancelled across the finite-node-count universe, leaving residual positive constant-stress that drives persistent edge-length-growth: dark-energy. In-framework formula for dark-energy-density:

$$\rho_\Lambda = \frac{\alpha}{r_B^2 \cdot N_0^{2/3}} \cdot f_{\text{closed-shell}}$$

Where α = fine-structure-constant, r_B = Bohr-radius, N_0 = total-node-count, $f_{\text{closed-shell}}$ = closed-shell geometric-factor.

Dark-matter = background of non-standing-wave travelling-modes across whole-network

All nodes not forming closed-standing-wave-configurations compose a travelling-wave-background. It influences visible-matter-kinematics via geometric-stress without coupling to electromagnetic-radiation.

Cosmic-microwave-background (CMB) = relics of global-network

early-vibrational-modes

In the early-universe all-nodes vibrated with high-amplitude producing global standing-wave-patterns. As networks-cool (average-amplitude decreases), these patterns freeze-in and survive as CMB-radiation. Temperature-fluctuations of CMB correspond to tiny primordial-perturbations in node-density and phase.

Cosmic periodic-oscillation

Expansion is not eternal. Once average-edge-length reaches maximum-value $L_{\max}$, geometric-stress dominates triggering global-network-re-configuration: edge-lengths contract (shrinking-half-cycle), shrink to minimal-value and rebound initiating a new expansion-cycle. The universe undergoes periodic oscillation: expand-reconfigure-contract-rebound.

Phenomena explainable in this subsection:

- Hubble-expansion: growth of network-average-edge-length.
- Dark-energy: global residual phase-strain of whole-network.
- Dark-matter: free-nodes not organized into standing-wave-configurations.
- Cosmic-microwave-background: frozen-remnants of global-early-vibrational-modes.
- Cosmic periodicity: expansion-contraction stress-cycles.

4.11 Systems: emergence of self-replicating units and intelligence

Definition of a System

Within this paper a system means emergent objects capable of recognising

environmental-states and actively taking actions to sustain their own existence.

System-categories contain three high-order emergent-structures:

1. Natural biological-systems;
2. Human-built artificial-intelligence and information-systems;
3. Social-systems assembled by living-individuals, including enterprises, organisations, societies and nation-states.

Differing from purely-physical passive-equilibrium-objects, systems feature teleology oriented toward self-preservation. Their internal-interaction-modes are no longer passively determined exclusively by underlying-physical-rules. They rely upon sensory-perception and feedback-regulation to continuously maintain structural-boundaries and prolong their own persistence.

Emergence of self-replicating-units: critical-threshold for system-birth

Within node-network cosmic-evolution purely-physical-structures can only achieve mechanical-thermodynamic-equilibrium and possess no teleological self-preservation. The birth of systems occurs at one critical bifurcation: emergence of self-replicating replicators. Two-level key-emergences exist within cosmic-evolution:

1. Chemical-level: emergence of self-replicating composite-molecular-systems providing material-pre-conditions for biological-systems.
2. Biological-level: living-beings evolve perception-recognition-decision-making-abilities and construct artificial-logic-units, laying evolutionary-pre-conditions for intelligent-systems.

Both biological-genetic-systems and man-made computing-systems share minimal-operable evolutionary-logic-architectures built-from

complementary-binary-structures:

System-type	Complementary binary-structure
Biological inheritance	Base-pairs (A-T/U, G-C)
Computation	0 / 1 plus logic-gates

Spontaneously-emergent minimal self-replicating-systems must contain at-least two mutually-recognisable complementary monomer-species X and Y. Different-sequences such as XYXYX or XXYYX fold into distinct spatial-geometries exhibiting differential catalytic-activity. Structural-differences yield functional-differences; functions can be filtered-by-environment, heritable and mutable — establishing prerequisites for evolution by natural-selection.

Primitive replicators are linear heteropolymer-structures: using their own-chain as template, capturing activated environmental-monomers to synthesise complementary-strands. Environmental-thermal-perturbations drive duplex-unzipping and new-strand-detachment completing one full self-replication-cycle.

Primitive logical boundary between self and non-self

After emergence of self-replicating-units the universe obtains its first functional logical boundary:

- Self: template-chain-structure acting as replication-template.
- Non-self: environmental free-monomers, alternative-chain-structures and outer-environment.

This boundary arises from purely-physical-mechanisms and does not depend on subjective-consciousness or subjective-sensation, comparable to mechanical-devices distinguishing-objects purely-via shape-and-charge-matching-criteria. It marks the transition-point of the universe from undifferentiated passive-physical-equilibrium toward bounded, persistence-oriented system-structures, and forms the archetypal origin-point for all life, intelligence and cognition.

Through iterative-evolution of self-replicating-structures simple-life evolves sophisticated perception-homeostasis-adaptable-behaviour eventually yielding human-level higher-intelligence. Humans in-turn build artificial-information-systems and computational-intelligence-systems simulating cosmic-laws, closing the highest-order cognitive-loop of cosmic-hierarchy.

General constructive-rules for systems: recursive full-life-cycle-architecture

All systems as persistence-oriented high-order-entities obey full-life-cycle-evolution-rules: general-phases include construction, operation and destruction; complex-systems may further-subdivide phases. The ultimate-purpose of all internal-sub-objects within a-system is to serve overall-system-persistence: core-sub-objects deliver main-functions; auxiliary-sub-objects implement environmental-adaptation, input-output, homeostatic-support and mode-switching-services. These constraints are recursive across systems, subsystems and components. Recursive-rules support system-structural-analysis, necessity-validation, redundancy-optimisation and forward-architectural-design.

Algorithms for full-life-cycles and system-engineering-construction belong to intermediate-layer general-cognition-and-engineering-implementation of this framework. This paper primarily addresses N2T-physical-model and global-unified-modelling-paradigm; further-details on system-engineering, algorithms and artificial-system-implementation will not be expanded-upon here.

5 Summary of Derivations: Derivation of 20+ Physical Quantities and Emergent Properties at Each Hierarchy

Within this framework, more than twenty physical quantities emerge naturally from the three preset parameters without introducing independent presuppositions. The following table lists these derived quantities, their corresponding observational values and their framework-level origins.

Derived Quantities

Category	Emergent Quantity	Corresponding Observed Value	Origin within the Framework
Fundamental Constants	Speed of light c	$2.9979 \times 10^8 \text{ m/s}$	$c = d_0 \cdot \omega_0$, combined from preset parameters
	Reduced Planck constant $\hbar$	$1.054 \times 10^{-34} \text{ J} \cdot \text{s}$	$\hbar = \omega_0 \cdot d_0^2 \cdot \alpha$, combined from preset parameters
	Gravitational constant G	$6.674 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$	Macro-statistical m^2/kg^2 of geometric stress
	Planck length l_p	$1.616 \times 10^{-35} \text{ m}$	$l_p = d_0 \cdot \alpha^2$, derived from preset parameters
	Elementary charge e	$1.602 \times 10^{-19} \text{ C}$	Divergence of phase stress S_θ
Particles	Electron mass m_e	$9.109 \times 10^{-31} \text{ kg}$	Energy of hexagonal-loop standing-wave (calibrated value, determined by

Category	Emergent Quantity	Corresponding Observed Value	Origin within the Framework
			framework topology)
	Proton mass	$1.672 \times 10^{-27} \text{ kg}$	Standing-wave modes of shared-node triple-quark rings
	Neutron mass	$1.675 \times 10^{-27} \text{ kg}$	Standing-wave modes of triple-quark rings (slightly heavier than proton)
	Electron spin	$1/2$	Topological phase $\Theta = \pi$ of hexagonal loop
	Quark colour charge	Three colours	$0^\circ, 120^\circ, 240^\circ$ phase differences of triangular loops
Atoms	Bohr radius r_B	$5.291 \times 10^{-11} \text{ m}$	Standing-wave condition of outer-shell grids (calibrated and

Category	Emergent Quantity	Corresponding Observed Value	Origin within the Framework
			consistent with d_0)
	Periodic table of elements	Observational data	Sequential expansion of outer-shell-grid topology with increasing Z
Nuclei	Nuclear radius formula	$R \approx 1.2 A^{1/3} \text{ fm}$	Macro-statistical average of geometric stress S_g
	Magic numbers (nuclear shell-closure)	2, 8, 20, 28, 50, 82, 126	Ratio of inner-to-outer stress σ_{in}/σ_{out} falling within interval $[1 - \alpha/2, 1 + \alpha/2]$
Stars	Stellar minimum mass	$\approx 0.08 \sim 0.1 M_\odot$	Intersection of resonance condition $N_{shared} \approx 137$

Category	Emergent Quantity	Corresponding Observed Value	Origin within the Framework
			and gravitational confinement
Cosmology	Hubble constant H_0	$\approx 70 \text{ km/s/Mpc}$	Relative growth rate of network-wide average edge length
	Dark-energy density ρ_Λ	$\approx 10^{-9} \text{ J/m}^3$	Global residual phase-strain; combined by α, r_B, N_0
	CMB temperature	2.725 K	Frozen relic of early global network vibrational modes
	Vacuum-energy density	$\approx 10^{-9} \text{ J/m}^3$	Sum of zero-point energies for finite node network, cut-off by minimum stable distance

Emergence Manifested Across Hierarchies

Hierarchy	Underlying Mechanism	Resultant Emergent Properties
Fundamental nodes	Receive, Update, Transmit (three basic rules)	Vibrational waves, phase differences, stress
Quarks	Three-node locking at 120° phase offset	Colour charge, colour confinement, mass
Atoms	Phase-locking between electron-shell standing-waves and atomic nuclei	Periodic table, orbital shapes, chemical-bond directions
Stars	Outer-shell grid sharing and resonance amplification	Nuclear fusion, element synthesis, energy release
Galaxies	Phase differences between shared-nodes and free-nodes	Galaxy rotation curves, dark-matter distribution, large-scale cosmic structures
Cosmos-scale network	Periodic fluctuation of network-wide average strain	Hubble expansion, dark-energy, cyclic cosmic oscillation

Hierarchy	Underlying Mechanism	Resultant Emergent Properties
Systems	Emergence and recursive iteration of self-replicating units	Life, perception, intelligence and social structures

In every layer, macroscopic properties are not pre-designed, but generated by the operation of internal mechanisms.

6 Open-Problems Remaining Unsolved

Although this framework is self-consistent, several key topics still require deeper mathematical formalization:

1. Exact correspondence between topological configurations and symmetries of the Standard Model

We have mapped triangular-loops to quarks (SU(3)) and open-chains to neutrinos (SU(2) weak interaction). However, the rigorous Lie-group / network-topology isomorphism has not yet been established. Further derivation is required to explain why triangular-loops correspond precisely to SU(3), rather than SU(4) or SU(5).

2. Quantization conditions for standing-wave modes

We have proposed that standing-waves form particles, yet an exact quantization formula relating standing-wave frequencies to particle mass is still missing – that is, the discrete version of the energy-mass relation $E = \hbar\omega$. It remains to define how loop perimeter, edge-count and wave-speed determine particle mass-spectra.

3. **Emergence of time and causality**

At present the model is quasi-static: the network evolves toward stress-minimum via iteration. It has not yet clearly explained how the arrow of time (entropy-increase direction) emerges from underlying dynamics. Further work needs to introduce non-equilibrium evolution, phase-transition irreversible processes and initial-cosmological conditions.

4. **Network-theoretic correspondence for dark-matter and dark-energy**

Detailed investigation is still pending on whether dark-matter corresponds to some kind of non-standing-wave travelling-wave background, and whether dark-energy originates from uniform strain-expansion across the whole-network.

5. **Experimentally-verifiable predictions**

We need to formulate testable experimental or observational predictions, for example: cosmic standing-wave resonance signals in the low- l region of CMB power-spectra, or novel particle configurations / topological defects.

7 **Quantitative Verification: Two Key Calculations of the Framework**

This section briefly illustrates how the framework derives two observational values starting from its underlying parameters. These results traditionally require extra ad-hoc assumptions.

Both calculations share common features:

- They are derived purely from the three fundamental parameters: total node count N_0 , initial average spacing d_0 , initial vibration frequency ω_0 .
- No additional tuning parameters or hypothetical mechanisms are introduced.

- The derived results agree with observations on the order-of-magnitude; partial numerical results show close agreement (about **20** deviation for stellar minimum-mass, and the vacuum-energy discrepancy can be reduced within one order-of-magnitude by geometric corrections).

7.1 Natural Cut-off for the Vacuum-Energy Catastrophe

Recall the dilemma in traditional quantum-field-theory: under the assumption of continuous space, infinitely many vibrational modes yield a theoretical vacuum-energy density of 10^{113} J/m^3 , while astronomical observations give about 10^{-9} J/m^3 . There exists a huge 10^{122} -fold discrepancy, known as the vacuum-energy catastrophe in physics.

Derivation within this framework:

Space is represented as a discrete node-network with finite total nodes $N_0 \approx 10^{80}$. Accordingly, the total number of accessible vibrational modes is finite. The highest effective frequency is truncated by the minimal stable distance:

$$l_{\min} = d_0 \cdot \alpha^2 \approx 2.82 \times 10^{-15} \text{ m}$$

$$\omega_{\max} = \frac{c}{l_{\min}} \approx 1.06 \times 10^{23} \text{ rad/s}$$

Zero-point energy per node is approximately $\frac{1}{2} \hbar \omega_{\max} \approx 5 \times 10^{-12} \text{ J}$.

Total zero-point energy across the observable universe:

$$E_{\text{vac}} \approx N_0 \cdot \frac{1}{2} \hbar \omega_{\text{max}} \approx 10^{68} \text{ J}$$

Distributed over observable-universe volume $V_{\text{obs}} \approx 10^{80} \text{ m}^3$, we obtain vacuum-energy density:

$$\rho_{\text{vac}} \approx \frac{E_{\text{vac}}}{V_{\text{obs}}} \approx 10^{-12} \text{ J/m}^3$$

After introducing the hexagonal closed-shell geometric correction factor (roughly $4\pi \times 137 \approx 1720$), the value can approach $\approx 10^{-9} \text{ J/m}^3$, consistent with observational order-of-magnitude.

Conclusion: Vacuum energy yields a finite value rather than infinity. No fine-tuning is required. The discrete-geometric natural truncation replaces the divergent summation arising from infinite degrees-of-freedom in quantum-field-theory, turning vacuum-energy from an incalculable infinity into a finite computable observable quantity.

7.2 Geometric Derivation of the Stellar Minimum-Mass

Problem review: Observed minimum stellar mass is around $0.08 M_{\odot}$. Conventional models require complex numerical simulations to reproduce this value.

Framework derivation:

- Resonance amplification: When outer-shell grids overlap, amplitude grows proportional to $\sqrt{N}$, while stress scales as N .

- Fusion threshold: Approximately $N_{\text{shared}} \approx 1/\alpha \approx 137$ hydrogen nuclei must participate simultaneously in shared-grid resonance.
- Gravitational confinement: About 10^{56} resonance units must be gravitationally trapped within the same spatial region.

Combining gravitational-confinement and resonance-condition yields the theoretical stellar minimum-mass:

$$M_{\text{min}} \approx 0.1 M_{\odot}$$

Conclusion: The stellar-mass minimum arises from the intersection between microscopic resonance requirements and macroscopic gravitational confinement. There is no need to postulate an ad-hoc critical-mass or initial-mass-function.

8 Testable Observational Predictions

Based on the core logic of the framework, the following test-able predictions are proposed:

Prediction 1: Additional small-amplitude periodic oscillations in specific angular scales of the CMB spectrum

Rationale: The cosmic network itself forms a finite closed structure. Collective vibrational modes of all nodes generate global standing waves. These macroscopic

standing-waves modulate the phase distribution on the last-scattering-surface and are superimposed onto standard acoustic oscillations.

- **Predicted signature:** A set of equally-spaced secondary peaks with small amplitude appear in the low- l region of the CMB spectrum ($l < 30$). Their amplitude is roughly $\alpha \approx 1/137$ relative to the main peak near $l \approx 200$.
- **Test method:** Analyze data from the Planck satellite and future observatories such as CMB-S4 and LiteBIRD, searching for deviations from pure Λ CDM standard-model predictions.
- **Falsification condition:** If no extra periodic signals consistent with the “global standing-wave” hypothesis are found in the low- l range, the framework needs revision or abandonment.

Prediction 2: Topological-soliton new-particles with fixed mass windows produced in high-energy particle collisions

Rationale: The fundamental underlying topology of the universe consists of three-node triangular-loops and six-node hexagonal-loops. Current accelerators only smash and recombine known composite topologies and have not yet reached the energy threshold for directly producing bare closed-loop solitons. When the center-of-mass collision energy reaches the range from hundreds of MeV up to several TeV, pure bare triangular-loop solitons and bare hexagonal-loop solitons without outer-grid confinement may be generated. These topological structures do

not belong to known Standard-Model particles; they possess stable intrinsic vibrational frequencies corresponding to fixed mass windows.

- **Predicted signature:**

- Bare triangular-loop soliton (primitive quark topology): mass concentrated in the **200– 800 MeV** range. It carries no complete electromagnetic phase-stress, exhibits only short-range strong-interaction coupling, and mainly decays into pairs of light quarks.
- Bare hexagonal-loop soliton (primitive electron topology): mass window of **0.4– 1.2 GeV**. It carries unit phase-divergence electric charge, spin-**1/2**, and cannot form atomic-orbital standing-waves without atomic-grid confinement.
- Both soliton types show characteristic decay signatures: no intermediate gluon shower transition, directly decaying into **2– 3** Standard-Model leptons / quarks. Their decay tracks can be clearly distinguished from ordinary resonance-states.

- **Test method:** Use high-luminosity LHC runs and the future FCC hadron / lepton collider. Screen low-background, non-standard-shower short-lived resonance events, and statistically analyze event mass distributions for isolated resonance peaks in the predicted mass windows.

- **Falsification condition:** If massive collision datasets accumulated over long running-times never show new soliton resonance peaks matching the predicted mass-range and decay signatures, core assumptions about particle closed-loop topology in this framework require major revision or rejection.

Prediction 3: Fixed-spectrum pulsed radiation from topological surface-defects on neutron stars, plus quantitative mass-magnetic-field relation for neutron stars

Rationale: Inside neutron stars, electron outer-shell grids are completely crushed. Triangular-loop neutron topologies are tightly stacked. During high-speed rotation, local topological misalignment defects appear at the stellar surface. Stress concentrates at defect positions and emits coherent phase-beams, generating pulsar periodic radiation. Meanwhile, spin-twist orientations of internal neutron triangular-loops become frozen and highly ordered during compression. The degree of spin ordering is determined by the number of shared edges, which directly sets the global magnetic-field strength of the neutron star.

- **Predicted signature:**

- Pulsar radiation spectra exhibit fixed segmented features: low-frequency components originate from collective neutron bulk-vibration; high-frequency components arise from local topological defects on the surface. The spectral inflection-point is uniquely determined by the average grid edge-length of the neutron star.
- Neutron-star surface magnetic-field and stellar mass follow a piece-wise monotonic positive power-law relation: $B \propto M^{3/2}$. Very few neutron stars should deviate significantly from this power-law.
- For massive neutron stars near the black-hole critical-mass, surface topological-defect density increases noticeably; pulse width of pulsed radiation narrows accordingly, yielding a clear mass-pulse-width correlation curve.

- **Test method:** Large-sample astronomical observations in X-ray and radio bands. Precisely measure mass, magnetic-field, radiation spectrum and pulse width for thousands of pulsars. Statistically fit the mass-magnetic-field

power-law, spectral inflection-point distribution and mass-pulse-width relation, and compare with quantitative model predictions.

- **Falsification condition:** Large-sample neutron-star statistics completely depart from the $B \propto M^{3/2}$ power-law, and no predicted unified spectral-inflection-point feature is observed. Then the inferences that “neutron-stars form by grid compression and magnetic-field comes from topological spin-ordering” become invalid.

Overall falsification logic: If all three observational signatures are absent from measurements, the whole framework must be discarded. Failure of a single prediction only triggers revision of partial topological assumptions.

9 Unified Modeling for All Objects

In this chapter, we classify objects in the universe and demonstrate that physical objects and human society share isomorphic structures and can be analyzed within one unified model.

9.1 Classification of Object Categories

All objects in the universe fall into three categories:

- **Pure-physical objects:** particles, atoms, inorganic molecules, macroscopic celestial bodies, inanimate material structures.
 - No environmental-recognition capability
 - No self-sustaining existential goals
 - No proactive behavioral strategies
 - All changes are passive topological responses

- **Quasi-systems:** simple lower-level life-forms, basic biological structures, general-purpose information-systems, embedded intelligent devices, automated equipment, etc.
 - Possess limited environmental-recognition ability
 - Have basic capabilities for steady-state maintenance and structural persistence
 - Feature simple boundaries, fixed response mechanisms and finite lifecycles
 - Lack complete autonomous goal-systems; no high-level evolution or proactive adaptive strategies
- **Complete complex systems:** individual human beings, as well as societies, enterprises and nation-states composed of human individuals.
 - High-level environmental-recognition, cognitive-judgment and trend-perception capabilities
 - Strong endogenous goals for self-preservation, expansion and stability maintenance
 - Multi-nested architectures, division-of-labor roles, transaction workflows, institutional norms and dynamic feedback with long-cycle evolution
 - Full behavioral repertoire including active adaptation, reform, game-playing and environmental transformation

9.2 Unified Modeling of All Objects

Every object possesses the complete six-layer structure:

1. **Component layer:** composed of physical objects.
2. **Architecture layer:** physical connections among components.

3. **Mechanism layer:** standardized combinations of interaction modes between components.
4. **Transaction layer:** one complete interaction among components.
5. **Event layer:** one concrete instance of a transaction.
6. **Information layer:** material carriers and their states-attributes transmitted through interactions.

Mechanisms support the formation of complete internal operation and external interaction patterns. Transactions can be driven by periodic rhythms or trigger-conditions; nesting and invocation are supported.

Passive six-layer structure for purely-physical objects

Particles, atoms, molecules, celestial bodies and cosmic structures naturally feature the six-layer architecture, yet all manifest as passive constraints, fixed mechanisms and no self-regulating or self-sustaining purposes:

1. **Component layer:** physical structures built from subordinate nodes, sub-structures and nested topologies, naturally generating containment- and inheritance-hierarchies.
2. **Architecture layer:** fixed arrangements for particle lattices, celestial orbits and network topologies, forming stable configurations.
3. **Mechanism layer:** gravitation, electromagnetism, quantum stress and phase-synchronization constraints constitute invariant operating-rules. These rules are not designed for self-maintenance; they are inevitable behaviors derived from topological rules.
4. **Transaction layer:** particle collisions, chemical combination-decay, orbital revolutions and energy-exchange represent objective transaction flows.

Transactions have no goals, strategies or choices, and exhibit no self-adaptation.

5. **Event layer:** phase jumps, energy-level transitions, collision perturbations and structural disintegrations count as objective events. Events occur without serving structural continuity and merely obey global stress-balance.
6. **Information layer:** particle phase, state, spin and energy-levels represent primary physical information. This information objectively exists, yet it is not used for environmental-recognition, decision-making or self-preservation.

Quasi-systems: weak initiative without complete self-sustaining strategies

In quasi-systems the six-layer structure begins to support limited adjustable mechanisms and responses:

- Preliminary environmental sensing and simple steady-state maintenance;
- Partial capacity to counteract external perturbations;
- But no top-level self-preservation goals, no proactive evolution, no intentional design or strategic behavior.

Complete complex systems: core distinctions for higher-order life, enterprises and societies

For genuine complex systems, attributes across all six layers are reversed:

- Architecture can be reconstructed;
- Mechanisms can adapt and self-regulate;
- Transactions can operate strategically;
- Events may be predicted, avoided or exploited;

- Information is used for environmental perception, decision-making and iterative evolution.

Operations across all layers serve one overarching objective: sustaining continuous existence, stability and further continuation of the whole system.

10 Conclusions and Outlook

10.1 Summary: From Three Parameters to One Universe

From the perspective of constructing the universe, this framework can hardly be simplified further. It is not a simplified version of a complex theory; instead, it is a theoretical framework for constructing cosmic complexity starting from minimal underlying assumptions.

Starting at the most fundamental level, the framework adopts only three preset parameters: total number of nodes N_0 , initial average spacing d_0 , and initial vibration frequency ω_0 , together with five observational calibration values. It builds a complete derivation framework covering from quarks to galaxies. The core logic of Node-Network Theory can be summarized in one sentence:

The universe is a node-network. Nodes generate waves through vibration; phase differences of waves drive stress; stress drives nodes to rearrange, forming all structures from elementary particles to galaxies.

No preset fundamental constants are introduced throughout this process. The speed of light c , reduced Planck constant $\hbar$, gravitational constant G , fine-structure

constant α , elementary charge e all emerge naturally from geometric combinations of the initial parameters.

Dimension	Standard Physics	This Framework
Fundamental entities	Matter, fields, particles	Nodes, edges, vibration
Space	Pre-existing stage	Geometric structure of the node-network itself
Time	Independent dimension	Function of the average vibration amplitude of the whole network
Forces	Four independent fundamental interactions	Projections of a unified stress function under different topologies
Fundamental constants	$G, \hbar, c, e, \alpha$ etc. are independently preset	All are emergent quantities
Free parameters	19 (Standard Model)	3

10.2 Outlook: Three Directions for Further Advancement

At present, Node-Network Theory remains a thought-experiment-level theoretical construction. Further progress should be pursued along three main directions:

1. **Mathematical formalization:** Convert qualitative descriptions of the framework into precise mathematical models. Define the Hamiltonian of the node-network, derive the continuum limit of the stress function, and establish exact correspondence with existing theories including quantum field theory and general relativity.
2. **Computational simulation:** Implement numerical simulation of the node-network on computers. Starting from the initial parameters, let the network evolve self-organizing under stress-minimization rules, and observe whether emergent structures from quarks to galaxies can be reproduced.
3. **Experimental testing:** Propose experimentally-testable predictions accessible to current-day technology.
 - Vacuum stress effects, e.g. the micro-deformation experiment of liquid boundaries inside vacuum chambers;
 - Search for extra cosmic standing-wave resonance signals in the low- l region of CMB spectra;
 - Verify structural predictions of this framework for superconductors and transparent conductors in material-science experiments.

10.3 Final Open-Ended Question

This framework points toward a deeper philosophical question: Why is the underlying structure of the universe exactly a node-network?

The answer to this question may forever lie beyond the scope of physics. It brings us back to our original observation: we study the universe using sensory organs and thinking capabilities that are themselves constructed from components of that very node-network universe.

References

1. Feynman, R. P., Leighton, R. B., & Sands, M. (1963). *The Feynman Lectures on Physics*. Addison-Wesley.
2. Griffiths, D. J. (2008). *Introduction to Elementary Particles* (2nd ed.). Wiley-VCH.
3. Weinberg, S. (1995). *The Quantum Theory of Fields* (Vol. 1-3). Cambridge University Press.
4. Misner, C. W., Thorne, K. S., & Wheeler, J. A. (1973). *Gravitation*. W. H. Freeman.
5. Planck Collaboration. (2020). "Planck 2018 results: VI. Cosmological parameters." *Astronomy & Astrophysics*, 641, A6.
6. Aaboud, M., et al. (ATLAS Collaboration). (2016). "Search for resonances in diphoton events at $\sqrt{s} = 13 \text{ TeV}$." *Physical Review Letters*.
7. Wheeler, J. A. (1990). "Information, Physics, Quantum: The Search for Links." In *Complexity, Entropy, and the Physics of Information*. Addison-Wesley.
8. Bostrom, N. (2003). "Are You Living in a Computer Simulation?" *Philosophical Quarterly*.

About the Author

Cunzhang: Systems Analyst, Software Designer, Senior Network Architect. Research interests: information meta-model, six-layer system meta-model, information-system framework.

Copyright Statement

This work is licensed under the Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0). When re-using this work, please provide attribution: Cunzhang – <http://114.55.33.140/archives/1122>

Revision History

- 2026-07-01: Initial draft started.
 - 2026-08-01: Completion of the main content.
 - Ongoing optimization...
-

CUNZHANG